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Keeping an eye on things

Module 8 - Automation
Google Cloud Computing Foundations

Course map

01 — So, what's the cloud anyway?

02 — Start with a solid platform

03 — Use Google Cloud to build your apps

04 — Where do I store this stuff?

05 — There's an API for that!

06 — You can't secure the cloud, right?

07 — It helps to network

08 — Keeping an eye on things

09 — You have the data, but what are you doing with it?

10 — Let machines do the work

Google Cloud skill badges

Objectives

01 Learn about Infrastructure as Code (IaC).

02 Explore Terraform as an IaC option.

03 Examine the role of monitoring, logging, error reporting, tracing, and profiling in the cloud.

04 Learn how to use Google Cloud operations suite for monitoring, logging, error reporting, tracing, and profiling.

Agenda

01 Introduction to Infrastructure as Code (IaC)

02 Terraform

03 Monitoring and managing services, applications, and infrastructure

04 Google Cloud's operations suite

05 Lab: Cloud Monitoring: Qwik Start

06 Quiz

07 Summary

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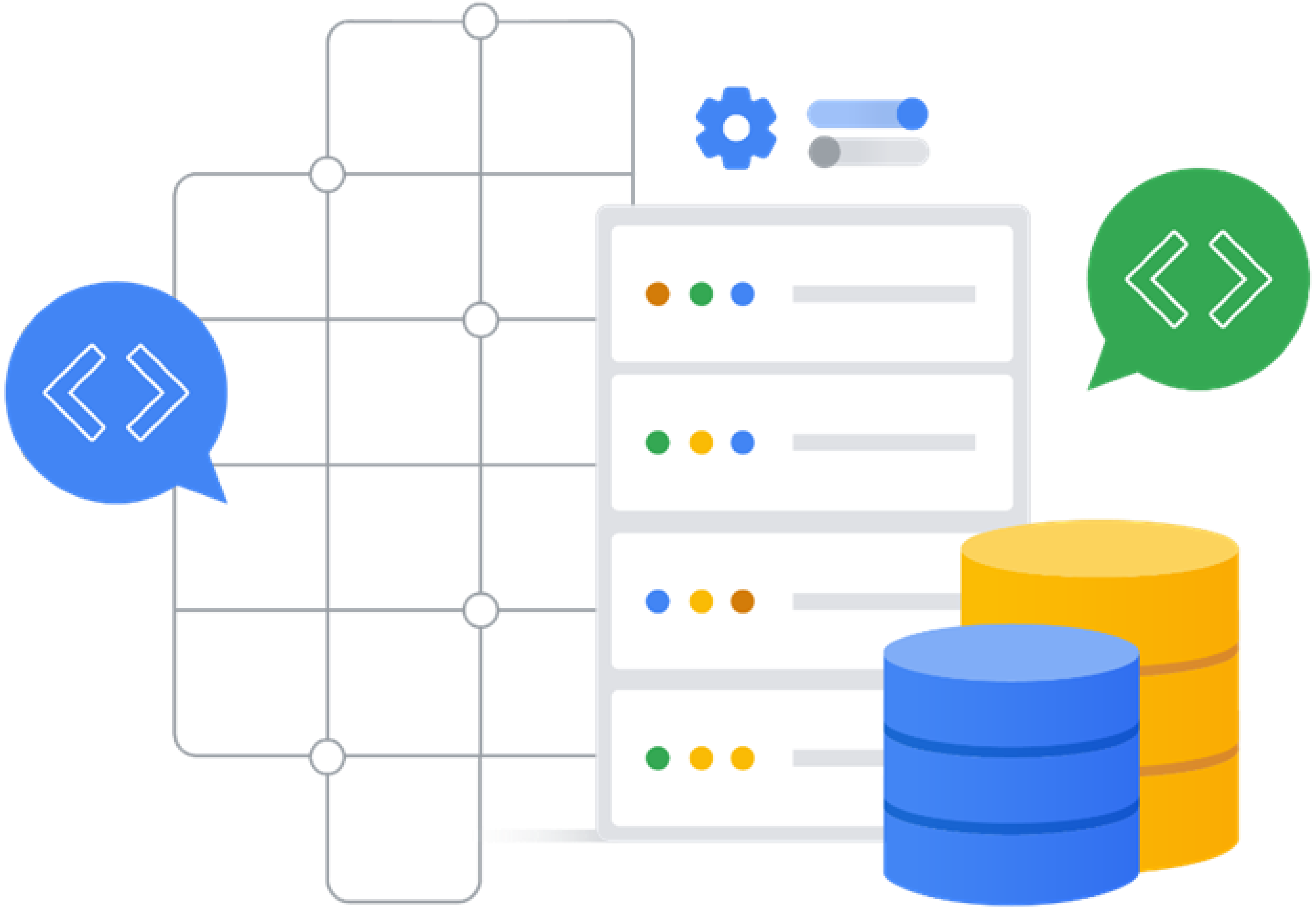
Manually set up an environment in Google Cloud

01 Updating commands if you want to change the environment

02 Writing new commands if you want to clone an environment



Define your required infrastructure as code

- Infrastructure requirements are defined as code in a template that's human readable and machine consumable.
- Use templates to automate building, modifying, and deleting infrastructure.
- Templates can be stored, versioned, and shared.
- Templates can be used to rebuild an infrastructure after a crash.

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01 Introduction to Infrastructure as Code (IaC)

02 **Terraform**

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Terraform can be used to provision resources

Template

Terraform templates describe your environment

- Create a template file using HashiCorp Configuration Language (HCL)
- The template determines the actions needed to create the environment
- A deployment can be repeated as many times as required
- Terraform uses the underlying APIs of each service to deploy your resources
- Store and version-control Terraform templates in Cloud Source Repositories

Google Cloud support is also available for third-party, open source IaC tools

Google Cloud

- Ansible
- CHEF
- Packer by HashiCorp
- Puppet

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Monitoring is the foundation of product reliability

- Reveals what needs urgent attention
- Shows trends in application usage patterns
- Helps improve an application experience

Monitoring gives you real-time system information

Google's *Site Reliability Engineering* book

landing.google.com/sre/books

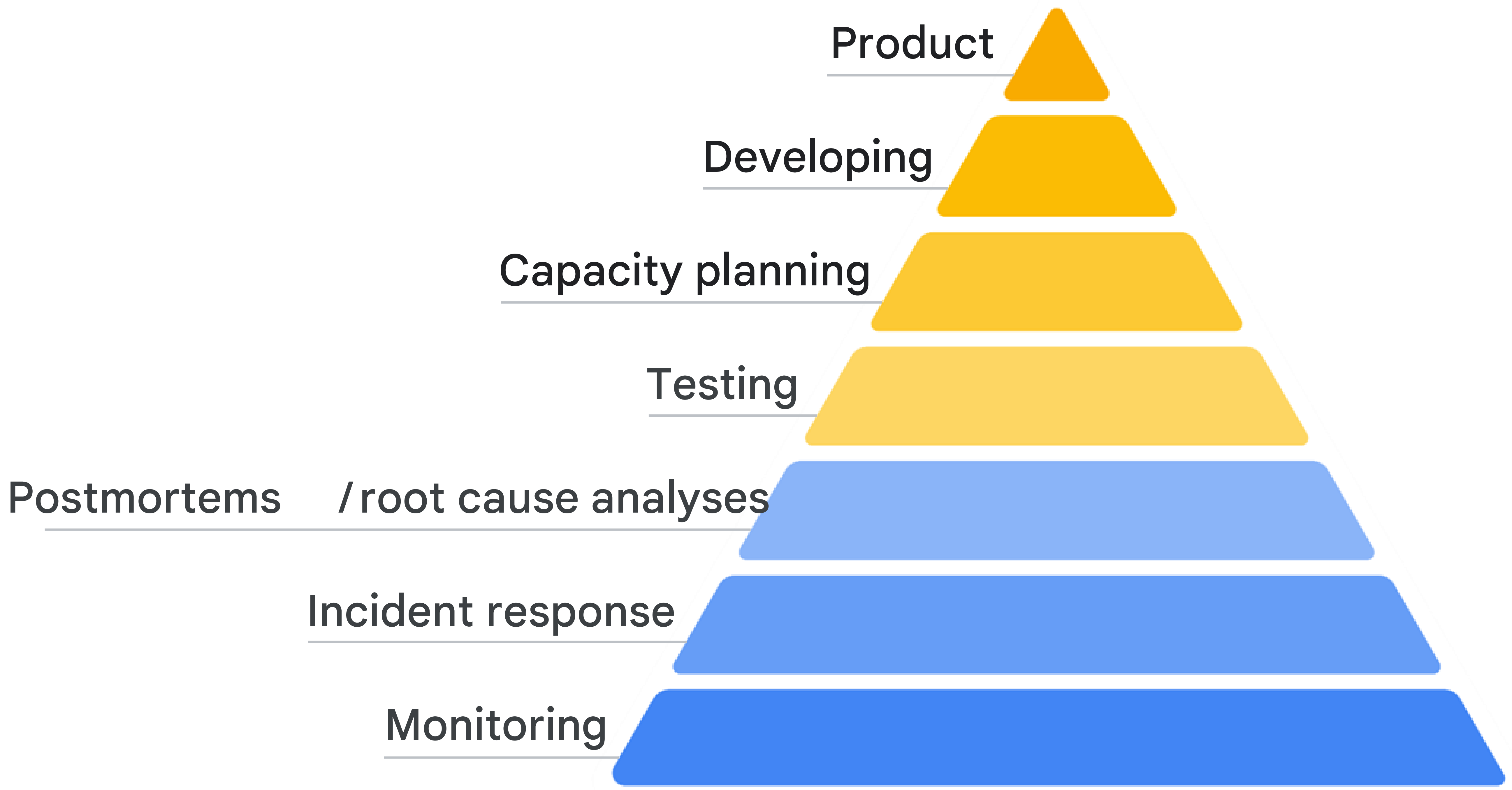
Collecting, processing, aggregating, and displaying real-time quantitative data about a system, such as:

- Query counts and types
- Error counts and types
- Processing times
- Server lifetimes

Benefits of monitoring

- Ensure continued system operations
- Uncover trend analyses over time
- Build dashboards
- Alert personnel when systems violate predefined SLOs
- Compare systems and systems changed
- Provide data for improved incident response

Monitoring is the foundation of product reliability



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Products for operations roles

Monitoring

Logging

Error reporting

Debugging

Monitoring

BigQuery

Queries in flight, scanned bytes billed, slots used

Compute

CPU and memory utilization, Uptime, disk throughput

Cloud Run

CPU utilization, billable time, memory utilization

Applications

OpenTelemetry custom metrics

Cloud Monitoring

- Provides visibility into the performance, uptime, and overall health of cloud-powered applications
- Collects metrics, events, and metadata from projects, logs, services, systems, agents, custom code, and various common application components
  - Includes Cassandra, Nginx, Apache Web Server, Elasticsearch, and many others.
- Ingests that data and generates insights via dashboards, Metrics Explorer charts, and automated alerts

Cloud Logging

Analyze
- Analyze log data in real time with the integrated Logs Explorer
- Analyze exported logs from Cloud Storage or BigQuery

Export
- Export to Cloud Storage, or Pub/Sub, or BigQuery
- Create logs-based metrics for augmented Monitoring

Retain
- Data access and service logs 30 days (configurable), and admin logs for 400 days
- Longer retention available in Cloud Storage or BigQuery

Key log categories

Cloud Audit Logs

- "Who did what, where?"
- Admin Activity
- Data Access
- System Event
- Access Transparency

Agent Logs

- Fluentd agent
- Common third-party applications
- System software

Network Logs

- VPC flow
- Firewall rules
- NAT gateway
- Load Balancer

Service Logs

- Standard Out / Error
- Created with API

Error Reporting

- Counts, analyzes, and aggregates the crashes in your running cloud services.
- Management interface displays the results with sorting and filtering capabilities.
- A dedicated view shows the error details: time chart, occurrences, affected user count, first- and last-seen dates, and a cleaned exception stack trace.
- Create alerts to receive notifications on new errors.

Cloud Trace

- Collects latency data from distributed applications and displays it in the Google Cloud console.
- Captures traces from applications deployed on App Engine, Compute Engine VMs, and Kubernetes Engine containers.
- Performance insights are provided in near-real time.
- Automatically analyzes all of your application's traces to generate in-depth latency reports to surface performance degradations.
- Continuously gathers and analyzes trace data to automatically identify recent changes to application performance.

Cloud Profiler

- Uses statistical techniques and extremely low-impact instrumentation that runs across all production application instances to provide a complete CPU and heap picture of an application.
- Allows developers to analyze applications running anywhere, including Google Cloud, other cloud platforms, or on-premises, with support for Java, Go, Python, and Node.js.
- Presents the call hierarchy and resource consumption of the relevant function in an interactive flame graph.

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Lab: Cloud Monitoring: Qwik Start

Hands-on lab

During this lab you'll:

1. Create a Compute Engine instance.
2. Add an Apache2 HTTP server to the instance.
3. Create an uptime check and alerting policy.
4. Create a dashboard and chart.
5. View logs.
6. Check the uptime check results and triggered alerts.

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Google Cloud

Knowledge check

Where can you store and version-control your Terraform templates?

A. Cloud Monitoring

B. Cloud Trace

C. Cloud Profiler

D. Cloud Source Repositories

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Knowledge check

Which tool ingests metrics, events, and metadata to generate insights through dashboards, Metrics Explorer charts, and automated alerts?

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Knowledge check

In Cloud Logging, what is the default log retention period for data access logs?

A. 30 days

B. 365 days

C. 400 days

D. 3650 days

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Knowledge check

Which one of the following provides access to logs created by developers deploying code to Google Cloud?

A. Cloud Audit Logs

B. Agent logs

C. Network logs

D. Service Logs

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